Home Reading 04 — Transient Response

Before Class

Preview Lecture 4

Read Lecture 4: Transient Response.

Textbook Reading

- Required: Ogata, System Dynamics, 4th ed., Section 8-2, pp. 384–385, and Section 8-3, pp. 388–390 (first- and second-order responses).
- Page numbers above are printed textbook pages. Use the Lecture slides for the remaining worked examples.

This Week Introduces

First-order and second-order responses, response specifications, model identification and dominant poles.

Tutorial Preparation

Read Tutorial 4 before class. Be ready to:

- Q1: Recognise standard response parameters and their limits.
- Q2: Identify a test rig from its force-step response.
- Q3: Compare transient-response plots and interpret their parameters.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- What do time constant, damping ratio and settling time describe?
- Why can two systems with the same final value have very different transient responses?

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